

Proposal Template Guide

Read this page before drafting the proposal.

Template-only page. For the submission PDF, change `\showguidancetrue` to `\showguidancefalse` in `metadata.tex`. This switch removes this page and the blue writing guidance.

Why the proposal is important

- **Alignment:** defines the problem, scope, deliverables, and success criteria for the group, stakeholders (supervisors), and teaching team.
- **Feasibility:** shows that the approach, schedule, resources, budget, and risk controls support completion.
- **Traceability:** connects requirements, specifications, and stakeholder objectives to the design, verification tests, and validation.

Formatting and submission

Page setup: letter paper, 1-inch margins, 12-point sans-serif body text, and 1.5 line spacing. Tables use 12 point font and may use single spacing.

Length: Sections 1–6 have an eight-page maximum. The title page, executive summary, contents, references, and appendices are excluded.

Figures and tables: number, caption, cite, and discuss each figure and table. Use appropriate units and significant digits. Choose relevant diagrams appropriately and ensure they convey your message with enough detail, using context provided in the body of the report.

Final check: remove guidance; replace bracketed fields, placeholders, and the sample logo; update citations; confirm the page count; spell-check; and submit on onQ.

Systems engineering in this proposal

The following definitions from the *International Council on Systems Engineering (INCOSE) Systems Engineering Handbook*, published by the [INCOSE](#), convey the systems engineering (SE) perspective used in this proposal.

Systems engineering.

SE is an interdisciplinary approach and means to enable the realization of successful systems. It focuses on defining customer needs and required functionality early in the development cycle, documenting requirements, and then proceeding with design synthesis and system validation while considering the complete problem: operations, cost and schedule, performance, training and support, test, manufacturing, and disposal. SE considers both the business and the technical needs of all customers with the goal of providing a quality product that meets the user needs.

Systems engineering effort.

SE effort integrates multiple disciplines and specialty groups into a set of activities that proceed from concept to production and operation. SE considers both the business and the technical needs of all stakeholders with the goal of providing a quality system that meets their needs.

International Council on Systems Engineering [1, Sec. 1.5]

How the proposal is graded

The proposal will be graded with an eye towards model-based systems engineering (MBSE): diagrams, figures, and tables should convey technical information. The body of the text should introduce ideas, state their purpose, assumptions, and limitations, provide reasoning for decisions, and interpret the information shown. More generally, the proposal is graded based on appropriately communicating engineering design, using the rubric on the next page.



Logo example: `figures/sample_logo.png` is a generic sample. Set `\ProjectLogo` in `metadata.tex` to the team's image path, or leave it empty for a cover without a logo.

Dimension	Grading evidence
Problem, objectives, and scope	Stakeholder, system-context, use-case, objective, or requirements models define the need, system boundary, actors, operating environment, measurable objectives, scope, assumptions, and constraints. The executive summary and introduction explain the problem / motivation, project trajectory, deliverables, and how the proposed end product addresses the identified problem.
Functional description and traceability	Functional-decomposition, system block, interface, information-flow, signal-flow, state, activity, or sequence diagrams communicate system behaviour and structure. Consistent identifiers trace stakeholder needs → requirements → functions → components and interfaces. The text adequately explains these models.
Design, production, resources, and feasibility	Physical and logical architectures, function-to-component allocation models, trade-study tables, dependency networks, work-breakdown structures, milestone or Gantt diagrams, risk models, and resource or budget models show an implementable approach. The text justifies design choices, calculations, production methods, resources, budget, division of labour, contingencies, and the sequence of three to five credible milestones.
Verification and validation	Requirement traceability matrix, test flow charts, experimental test setup figures, timing or sequence diagrams, and / or other figures show how evidence progresses from components to subsystems and the integrated system. The text describes each prototype, states test methods, equipment or software, conditions, measurable acceptance criteria, required evidence, and why the planned tests sufficiently verify requirements and validate stakeholder needs.
Writing, formatting, and references	Every diagram has a descriptive caption, readable labels, defined symbols, units, directions, interfaces, requirement identifiers, and cited sources where applicable. There is consistency in terminology and identifiers. The title page, executive summary, contents, formatting, citations, and references are complete, clear, and professionally presented.

SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Proposal

Computer-Vision Autonomous Golf Cart

Fall 2026 / Winter 2027

WRITING TEMPLATE

Submitted by Group 07

Group member	NetID	Student number
Vince Song	21vs51	20341597
Connor Gorman	21crg6	20347278
Jp Masotti	21jpm28	20349226

Instructor: Prof. Muhammad Alam

Instructor email: m.alam@queensu.ca

Teaching assistant: Dimitria Silveria

TA email: lkx1@queensu.ca

Faculty supervisor(s): Karen Rudie

Supervisor email: karen.rudie@queensu.ca

Submission date: October 7th, 2026

Version: Version

Template setup: replace the bracketed fields and develop each section into a written project proposal. Include the industry partner when applicable.

Executive Summary

Aim for approximately three-quarters of a page. Summarize the project, the contents of this proposal, two or three main milestones, and the estimated total cost. Write this section after drafting the body.

[Paragraph 1: Identify the problem, intended users or market, proposed system, and expected benefit. State the project goal and the end product.]

[Paragraph 2: Summarize the principal functions and performance targets, the proposed design and production approach, and two or three milestones.]

[Paragraph 3: List the prototypes. Summarize the testing approach, resources and total estimated cost in the stated currency, and readiness to begin the project. If further research and / or learning is necessary, list the topics and their applicability.]

It's not necessary to follow the structure above. This is just a recommendation.

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Acronyms

ASCC	Autonomous Smart Caddy Cart
GPS	Global Positioning System
INCOSE	International Council on Systems Engineering
MBSE	model-based systems engineering
SE	systems engineering

1. Introduction

The body of the completed proposal, i.e. Sections 1–6 below, has a maximum length of eight pages; the title page, executive summary, contents, references, and appendices are not included in this limit.

In the intro below, establish the wider problem you are trying to address, project's scope in the context of this problem, the general design, and limitations.

Change the titles of these subsections as you see fit. We have just provided some example placeholders.

1.1 The shortfalls of autonomous golf carts

Walking golfers can already buy electric push carts that follow or lead the way on the course [2]. So, the problem of carrying your own clubs has already been solved, but where these carts fall short is the ability to actually guide how a golfer plays the game. The carts can't act like a human caddy would, which is where the purpose of the Autonomous Smart Caddy Cart (ASCC) comes in. Most golfers either track their information on paper, get info from their watch, phone, or a range finder [3], [4], [5]. The ASCC is an all in one solution to both club carrying, and caddy assistance. A golfer would usually need to manually input, or accept their distance, club information on their phone or watch. The ideal user of this cart is a golfer who wants to track their information, stay off their phone, and have a cheaper solution than the cost of a human caddy and electric cart combined.

Describe the problem, the intended end user or market, evidence of the need, and the value of addressing it. Support the context with several citations from peer-reviewed literature. Feel free to change the section titles to fit your project. This section is the 'Problem and Purpose' section.

1.2 A cost effective, phone-free autonomous smart caddy

The objective of this project is to design and prototype a simple tracking golf caddy that follows the golfer safely, identifies club usage without any input, and gives personalized advice to the golfer through a display and or speaker. Each outcome is measurable. The cart must hold its following distance (PF-03), respond to obstacles (PF-05), and detect tagged clubs (IF-04) within the targets in Table 3, and its shot-distance estimates and missing-club warnings are checked by the tests in Section 4. The engineering involved

is motor control, embedded sensing, golfer tracking, Global Positioning System (GPS) processing, and interface integration. Figure 1 shows the system in its operating context.

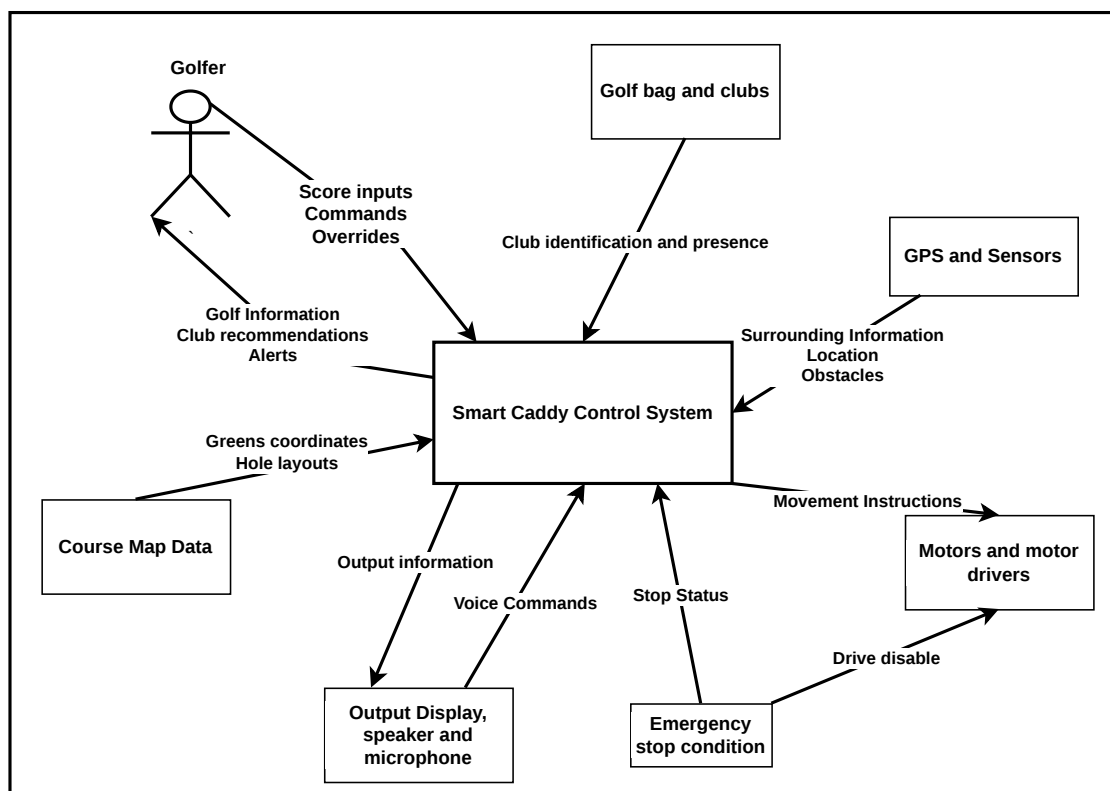


Figure 1: System context diagram for the Smart Caddy control system, showing the external entities it interacts with and the information exchanged with each.

Person-following robots track their target with cameras, laser rangefinders, or radio ranging, and the choice trades cost against robustness to lighting, occlusion, and bystanders [6]. Commercial following carts avoid the perception problem by tracking a handset that the golfer carries [2]. For shot tracking, Hall et al. tested two club-tag GPS systems against laser and tape measurements and found typical distance errors of three yards or less, but one system missed 67 shots where the other missed 3 [5]. Detecting the shot event is therefore the harder problem, not measuring its length.

The ASCC differs from these systems in where it senses. It detects a club leaving the bag instead of the swing, and it measures from the cart's position instead of the golfer's. This removes the phone and watch, at the cost of two errors that the test plan must quantify: a club removal that is not a shot, and the offset between the cart and the ball. Table 1 compares the three designs feature by feature.

Table 1: Comparison of the proposed ASCC design with existing following-cart and shot-tracking products.

Design consideration	Stewart Golf Q Follow [2]	Arccos Smart Sensors with Apple Watch [3], [4]	Proposed ASCC design
Equipment transport	Motorized cart carries the golf bag	Separate bag-transport solution is required	Motorized cart carries the bag and the golf-assistance hardware
Golfer following	Follows a carried Bluetooth handset; also supports remote control	No powered following hardware	Tracks one golfer and maintains a specified following distance
Club-use detection	None	Club-mounted sensors identify the club used on detected shots	Bag-mounted reader detects tagged-club removal and return
Shot distance and advice	Requires separate equipment	Associates shots with GPS locations; watch recommends clubs	Estimates distances from cart GPS locations at club events; onboard interface recommends clubs

State the specific objectives and measurable outcomes that address the problem. Characterize the electrical and / or computer engineering involved. Support your claims by describing similar approaches in the peer-reviewed literature. Include a general diagram showing the system context. Compare your approach to those in the peer-reviewed literature. This section is the 'Project Objective' section.

1.3 Deliverables: follow-me mobility, smart club tracking, and onboard caddy assistance

The deliverable is a single working prototype cart, demonstrated outdoors on open, dry grass with one golfer and one bag of tagged clubs. Its inputs are the golfer's position, button commands, club tag readings, GPS fixes, and obstacle detections; its outputs are wheel motion, displayed information, and warnings. The subsystems meet at the club event: a removal triggers a position record, and the distance to the next removal becomes the previous shot's estimated length. Tagging clubs in this way is a strategy also used by commercial shot trackers [3]. Table 2 lists each deliverable with its assumptions and limitations.

Table 2: Project deliverables, operating assumptions, and limitations.

Deliverable	Included functionality	Operating assumptions / constraints	Limitations / exclusions
Following-cart subsystem	Powered transport, one-golfer following, basic obstacle stopping, manual override	Dry grass; defined terrain and speed limits; golfer detectable by the tracking system	No independent course navigation; restricted terrain capability
Smart-bag subsystem	Tagged-club identification, removal/return events, missing-club warnings	One bag of tagged clubs; reliable reads within the defined bag configuration	Club removal does not confirm a shot; multiple-club handling requires evaluation
Golf-assistance subsystem	Estimated shot distances, player-specific club history, club recommendations	Usable GPS; identifiable successive shot events; green coordinates loaded beforehand	GPS error and cart-to-ball offset affect distance estimates; reliable putt measurement excluded
Onboard interface	Display and physical controls for core tasks	Information and controls accessible without phone interaction during play	Voice interaction remains an extension unless committed as a requirement
Integrated powered prototype	One working cart demonstrated with all core subsystems	Battery sized for the specified demonstration; dry-weather testing	Waterproofing and full 18-hole endurance excluded

This is a refinement of Sec. 1.2. Define the end product or products, the operating context, system inputs and outputs, and the major components. Explain each component's purpose and how the components fit together. State the project boundaries and assumptions. Address several limitations of your approach. Provide citations in the peer-reviewed literature and / or textbooks that apply similar strategies. This section is the 'Scope and deliverables' section.

2. Functional Description

2.1 System overview

Use a set of diagrams to explain the system boundary, internal functions, operating behavior, inputs and outputs, signal types, protocols, direction of information flow, computing architecture, and / or anything else that applies. Write a short paragraph here that explains design decisions and references each figure, like Figure 2. Maintain labeling consistency (such as IDs, components, units, etc.) across tables and figures in this document. It's a good idea to reference these later in Section 4.

Show the functional, physical, and logical diagrams.
Label components, inputs, outputs, arrows, and signal types or protocols.

Figure 2: [System architecture and information flow. (This is a generic example. Use several diagrams to show the various relationships between system components.)]

2.2 Interface and performance specifications

Give quantitative targets and tolerances with proper units. You can also use IDs to connect specifications with the tests in Section 4. Write a short paragraph here giving context for these requirements, citing the IDs as necessary.

The interface and performance specifications defined in Table 3 establish the measurable quantitative targets required to ensure the smart caddy effectively alleviates physical strain and cognitive load for senior golfers. Physical and hardware interfaces focus on ergonomic accessibility, payload capacity, and reliable user tracking (**IF-01** through **IF-04**). Performance metrics guarantee that the mechatronic chassis maintains safe following distances, handles standard 18-hole course topographies, and reliably detects active club removal and return (**PF-01** through **PF-05**). Each specification is tagged with a unique identifier to facilitate downstream verification and validation in Section 4.

Table 3: Preliminary system specifications.

ID	Requirement	Target	Tolerance
IF-01	User Tracking Beacon Range	15.0 m	± 2.0 m
IF-02	Onboard Display Diagonal Size	7.0 in	Minimum 5.0 in
IF-03	Bag Carrier Weight Capacity	20.0 kg	+5.0 / -0.0 kg
IF-04	Club RFID Detection Range	15.0 cm	± 5.0 cm
PF-01	Continuous Operating Range (Single Charge)	12.0 km	± 1.5 km
PF-02	Maximum Autonomous Cruise Speed	5.0 km/h	± 0.5 km/h
PF-03	Target Following Distance	1.5 m	± 0.3 m
PF-04	Maximum Incline In-Motion Gradeability	15.0 deg	Minimum 12.0 deg
PF-05	Obstacle Detection Response Delay	200 ms	≤ 300 ms

2.3 Risks and safety hazards

Identify relevant technical risks, health risks, and / or safety hazards, and indicate how you will manage these risks, and / or explain how system features and operating conditions are low-risk.

3. Design and Production Approach

3.1 Retroencabulator development

The autonomous golf pushcart will be developed through system design, subsystem prototyping, integration, testing, and refinement. The team has already obtained some technical components and will attempt to purchase a used motorized golf cart as the base platform. The Raspberry Pi, sensors, motor-control electronics, and smart-bag system will be integrated onto the cart, with development focused on motor control, sensor-based golfer following and obstacle detection, RFID-based club identification and tracking, and shot-data collection. A controller will be used to process sensor inputs and control the cart's movement during golfer following and obstacle avoidance. Each subsystem will be tested independently before being integrated into the complete system. Testing will begin in a controlled environment before progressing to outdoor conditions involving different walking speeds, obstacles, and terrain, with results used to refine the system and prioritize the MVP requirements.

3.2 Milestones and division of labor

*Plan at least five technical milestones. Give each milestone a date, an observable completion condition, and one responsible member as the primary lead. Course report and presentation deadlines do **not** belong here. This table tracks technical progress!*

Table 4: Technical milestones and ownership.

ID	Milestone	Due date	Lead
M1	Motorized control of the golf cart	October 30th, 2026	Connor
M2	Raspberry Pi integration and hardware intergration of the necessary sensors	November 20th, 2026	Vince
M3	RFID-Based Club Identification and Tracking	December 18th, 2026	Jp
M4	Obstacle detection/avoidance/emergency stop functionality	January 22nd, 2027	Vince
M5	User Interface Design and login system for golfer profile	February 5th, 2027	Jp
M6	Shot-Data Collection and Club Recommendation System	February 19th, 2027	Connor

M7	Full Subsystem Integration and System Refinement	March 1, 2027	All members
M8	Testing and validation of retroencabulator complete	March 15, 2027	All members

Add one or two explanatory sentences per milestone, and describe completion evidence and dependencies where applicable.

3.3 Contingencies

Identify likely technical or scheduling obstacles, the trigger for taking corrective action, any alternative approaches, and the expected effect on time and cost.

The main project risks include difficulty obtaining a suitable motorized cart, unreliable sensor-based golfer following, limitations in RFID detection, motor issues, and delays during system integration. If a component fails to meet requirements during testing, the team will modify the software or use an alternative sensor, controller, or RFID configuration. If RFID tags cannot be reliably detected within the golf bag, the team will test alternative identification tags such as NFC tags for example. If a suitable motorized golf cart cannot be obtained, a different cart or drive platform will be considered. Outdoor testing may be moved to a controlled environment if weather or terrain prevents reliable testing. If the project falls behind schedule, the team will prioritize the MVP features of basic cart movement, golfer following, obstacle detection, emergency stopping, and RFID-based club identification and tracking, while postponing more advanced features such as shot-based club recommendations or mobile-app functionality. These alternatives may add some cost or development time but will help ensure that the core system is completed.

4. Testing and Evaluation

Describe how the team expects to test the proposed system. The purpose of this section is to show that testing has been considered, not to provide a complete test procedure. (This you will do in the blueprint.) Use a diagram to communicate the overall test approach and briefly explain the expected evidence of success.

Insert diagrams showing the test environments, and link IDs to the requirements and specifications.
Show what will be tested, the main inputs and measurements, and how the results connect to the project requirements.

Figure 3: [Proposed system testing approach...be more specific here.]

Requirement	Proposed test	Expected evidence of success
[ID]	[Briefly describe the planned test.]	[State the expected result or pass criterion.]

List as many verification and validation tests as you can think of at this time. Briefly identify any equipment, software, facilities, or special conditions expected to be needed. Explain how the tests will help determine whether the final system meets its main requirements (verification) and stakeholder needs (validation).

5. Resource Requirements

5.1 Scheduling access to resources

Connect the technical milestones to laboratory space, equipment, fabrication, software, subscriptions, repositories, and other resources. Discuss access with the supervisor and include booking or lead-time requirements.

Table 5: Resource access by milestone.

Milestone	Resource / source	Access date / lead time
M1	[Tools and design resources]	[Date; lead time]
M2–M3	[Lab; equipment; fabrication]	[Date; booking]
M4	[Test equipment and facilities]	[Date; booking]

You can also relate each resource to the verification tests, or any other aspect of your report. Try to make use of labels / IDs so that they can be referenced consistently across the report.

5.2 Materials and other resources

Estimate costs in 600 CAD. Include quantities and the basis for estimates; identify resources supplied through existing access. Include tax, shipping, and contingency where applicable.

Table 6: Preliminary cost estimate (600 CAD).

Item / source	Qty.	Unit cost	Total
[Hardware / supplier]	[n]	[Cost]	[Cost]
[Software / service / access]	[n]	[Cost]	[Cost]
[Tax; shipping; contingency]			[Cost]
Estimated total			[Total]

[State the cost assumptions, access arrangements, and supervisor or industry contributions. Explain the funding and availability needed to complete the project.]

6. Conclusion

Summarize the project and the evidence established in Sections 1–5. End with the expected contribution and the team’s readiness to proceed.

[Identify the proposed system and its purpose. Recap the objectives, scope, principal specifications, design strategy, milestones, testing approach, resources, and estimated cost. Close with the expected benefit for the intended user or application.]

7. References

Replace these entries with the sources used in the proposal. Include previous project reports, papers, books, data sheets, and web resources as appropriate. Supply enough information to retrieve each source. Cite external claims, data, and adapted figures, and record permission or license details for reused material.

- [1] International Council on Systems Engineering, *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*, 4th ed., D. D. Walden, G. J. Roedler, K. J. Forsberg, R. D. Hamelin, and T. M. Shortell, Eds. Hoboken, NJ: John Wiley & Sons, 2015, ISBN: 978-1-118-99940-0.
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- [6] M. J. Islam, J. Hong, and J. Sattar, “Person-following by autonomous robots: A categorical overview,” *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1581–1618, Dec. 2019. DOI: 10.1177/0278364919881683